

Jiamin Li | Curriculum Vitae

601 Elm Avenue, Room 423 – Norman, OK 73019-3103

✉ jiamin.li@ou.edu • 📄 jli283.github.io

Employment

University of Oklahoma

Postdoctoral Associate

Mentor: András Lőrincz

Norman, OK

2024 – current

Education

University of Illinois at Chicago

PhD in Mathematics

Advisor: Wenliang Zhang

Chicago, IL

2018 – 2024

Binghamton University

BSc (Hons) in Mathematics

Binghamton, NY

2014 – 2017

Research Interests

Commutative algebra and algebraic geometry.

More specifically: local cohomology modules, asymptotic behaviors, multiplicities, D-modules and positive characteristics.

Papers

- Fourier transform of twisted equivariant D-modules. with András Lőrincz, in preparation
- On the generalized multiplicities of maximal minors and sub-maximal pfaffians, arXiv:2205.09657, preprint
- Singularities of generic linkage via Frobenius powers, Proc. Amer. Math. Soc. 152 (2024), no. 1, 53–61.
- Socle degrees for local cohomology modules of thickenings of maximal minors and sub-maximal Pfaffians, with Michael Perlman, Proc. Amer. Math. Soc. 152(2024), no.2, 599–615.
- Local cohomology and Segre products, with Wenliang Zhang, arXiv:2404.03469, preprint

Awards and Grants

2023 – 2024 Dean's Scholar Fellowship at the University of Illinois at Chicago

Winter 2023 JMM 2023 AWM Poster Award (funding to attend conferences)

Fall 2022 AMS graduate student travel grant

2017 Award for Academic excellence of Mathematics at Binghamton University

Talks

October 2025 50th Arkansas Spring Lecture Series, University of Arkansas

October 2024 Algebra and Representation theory seminar, University of Oklahoma

November 2023 Commutative algebra seminar, Purdue University

November 2023 Commutative algebra seminar, University of Michigan

November 2023 Algebra Days in Arizona, Arizona State University

October 2023 "Commutative Algebra, Differential Operators, and Singularities", AMS Sectional Meeting at Creighton University

June 2023 Lightning talk, Macaulay2 Workshop and Mini-school, University of Minnesota

October 2022 "Special session of commutative algebra", AMS Sectional Meeting at University of Utah

October 2022 Commutative Algebra and Algebraic Geometry seminar, University of Minnesota

March 2022 "Recent developments in commutative algebra", AMS Sectional Meeting at Purdue University

March 2022 Commutative algebra seminar, University of Illinois at Chicago

Posters

January 2023 JMM 2023 AWM poster session, Boston

October 2022 KUMUNU 2022 Fall conference (poster), University of Nebraska–Lincoln

Services

Co-Organizer of Special Session on Syzygies and Local Cohomology in Commutative Algebra
at the AMS Sectional Meeting at Tulane University Oct 2025

Co-Organizer of commutative and homological algebra student seminar at UIC 2023 – 2024

Co-Organizer of Graduate number theory seminar at UIC 2021 – 2022

Teaching Experience

at OU:

Fall 2025 Calculus III

Spring 2025 Linear Algebra

Fall 2024 Ordinary Differential Equations

at UIC:

Fall 2018 College Algebra

Spring 2019 Linear Algebra for Business

Fall 2019 Linear algebra for Business

Spring 2020 Linear algebra for Business

Fall 2020 College Algebra

Fall 2021 Calculus I

Spring 2022 Calculus II

Spring 2023 Linear Algebra for Business

Grader

Fall 2022 Applied linear algebra, Introduction to number theory

Spring 2021 Linear algebra, Abstract Algebra

Last updated February 28, 2026